

Software Architecture and Design
SOEN 343

Five Leaf Clover - HI System Phase I

Sofia Sanchez - 40247014 - Team Leader, Assumptions and Dependencies

Elizabeth Sinodinos - 40318356 - Product Definition, Technology Planned

Giuseppe Ippolito - 40315749 - Domain Model

Michael Santella - 40264501 - Context Diagram

Kailee Santucci - 40296552 - Product Perspective

Gina Cody School of Engineering and Computer Science
Concordia University

September 29th, 2026

Table of Contents

I- Problem Definition	
a. Problem Statement	1
b. Product Position Statement	2
II- Product Overview	
a. Product Perspective	3
b. Assumptions and dependencies	4
III- Technology planned: [technology used for the following activities]:	
a. Control version system	4
b. Team collaboration	4
c. Design and modelling work	4
d. Development framework	4-5
e. Coding	5
f. GenAI tools used	5
IV- Context diagram	5
V- Domain model	6
References	7

I- Problem Definition

Problem Statement

The problem of	Ordering food from restaurants using traditional methods (such as either calling the restaurant and placing a pickup or delivery order, or visiting the restaurant and ordering in person) is often tedious, time consuming and a deterrent to eating out. Additionally, these methods do not foster discovery of local restaurants and alternative cuisines. Another core issue is the difficulty in finding restaurants that tailor to your dietary requirements. Online research can incorrectly identify restaurants tailoring to a specific diet, or can miss some altogether.
Affects	This problem does not only affect customers who want convenient meal delivery and exposure to new restaurants, but restaurant owners who want to increase their exposure and improve sales. In addition, customers with dietary restrictions must overcome the difficulties in finding restaurants that meet their needs. Delivery drivers and couriers are affected as well since they are often limited to working for a single restaurant, which can decrease their revenue in times of reduced delivery orders.
The impact of which is	Longer wait times and lesser restaurant diversity for customers, decreased revenue and exposure for restaurants, decreased tips and revenue for delivery drivers, incorrect order fulfillment, and limited visibility of diet specific restaurants to customers.
A successful solution would	Enable convenient and fast food delivery for customers, restrict the proposed restaurants to those matching the dietary restrictions (in our solution, the focus will solely be on vegan food), increase exploration of cuisines, increase visibility and revenue for numerous restaurants, increase profit for delivery drivers and improve coordination of order placement + payment + order fulfillment + delivery.

Product Position Statement

For	The general public who are vegan and wanting food delivery, restaurants/restaurant owners, independent couriers
Who	Want easy and efficient food delivery, an application dedicated solely to their vegan diet, improved visibility and sales of their restaurant, and couriers wanting consistent and available deliveries to improve their wage.
The Five Leaf Clover - HI	Is a food delivery service web application specializing in vegan only food options.
That	Centralizes vegan restaurants to enable customers to discover and order meals without the hassle of researching and contacting restaurants directly, allows restaurants to advertise their menu to a larger audience and improve sales, and enables delivery drivers to increase their workload and profits.
Unlike	UberEats or DoorDash that focus on all types of restaurants and diets, or on grocery delivery.
Our product	Centralizes all restaurants that focus on vegan food, filtering out restaurants that do not meet this criteria. This simplifies food ordering and delivery for the customer, yet still benefits restaurants and couriers.

II- Product Overview

Product Perspective

The app is a web-based application for a food delivery system. The system allows users/customers to discover restaurants, browse menus, place orders, make payments, and receive order and delivery updates. Restaurants interact with the system to provide menus and manage incoming orders. Couriers use the system to receive delivery instructions and provide delivery updates throughout the process. The app is not entirely self-contained as it relies on external services to support some of its functions. An external payment processing service is required for processing payments. Additionally, mapping and location services (ex Google API) will provide location information used by the platform. The system is essentially an interface that coordinates information between customers, restaurants, couriers, platform operators, and external services.

Assumptions and dependencies

Assumptions	Dependencies
The restaurants provide accurate and up to date vegan menus, prices and availability	The app depends on info the restaurant provides to display available food options
Customers provide valid delivery address and valid payment info when they place an order	The app depends on external payment processing to get the customer's payment.
Delivery drivers are available to accept and complete delivery requests	Order delivery depends on driver availability and on the driver's ability to get to both the restaurant and the customer
Restaurants will receive and respond to orders placed through the app	Order fulfilment depends on restaurants accepting and preparing orders
Customers will have access to a web browser and internet	Customers need access to the internet to be able to use our application

III- Technology planned

Control version system

Github will be used as the version control system and code sharing for our application.

Team collaboration

The team will communicate and collaborate on all aspects of the project during lab time and by an iMessage group chat. Tasks will be divided equally between members and assigned using Github Issues. Github will be used to share our codebase between ourselves, the teaching assistant and professor.

Design and modelling work

Draw.io will be used to produce all diagrams found within the documentation for the application.

Development framework

Django is a high-level backend web programming framework that allows for fast, secure and scalable application development [1]. It was created to accelerate common web features, such as authentication, generating UI templates, and interacting with databases using Python. Next.js is a

full-stack framework that will be utilized for its built-in UI/UX optimizations, routing, and client-side interactions [2].

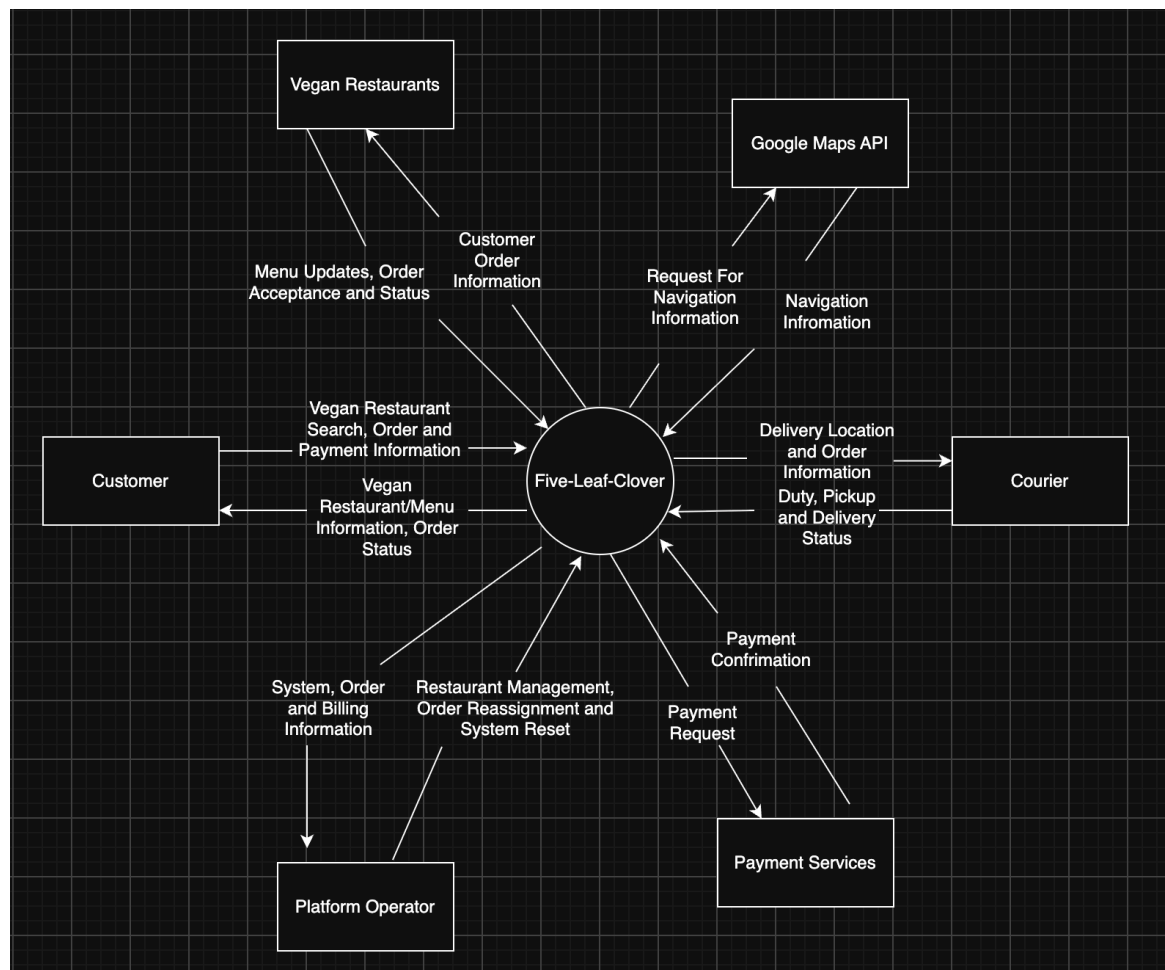
Coding

Python will be used for the backend development and business logic of our application [3]. It is an object-oriented language that is easy to read and that integrates well with our frontend development framework. This language will enable fast development of our application and maintainability as more features get added.

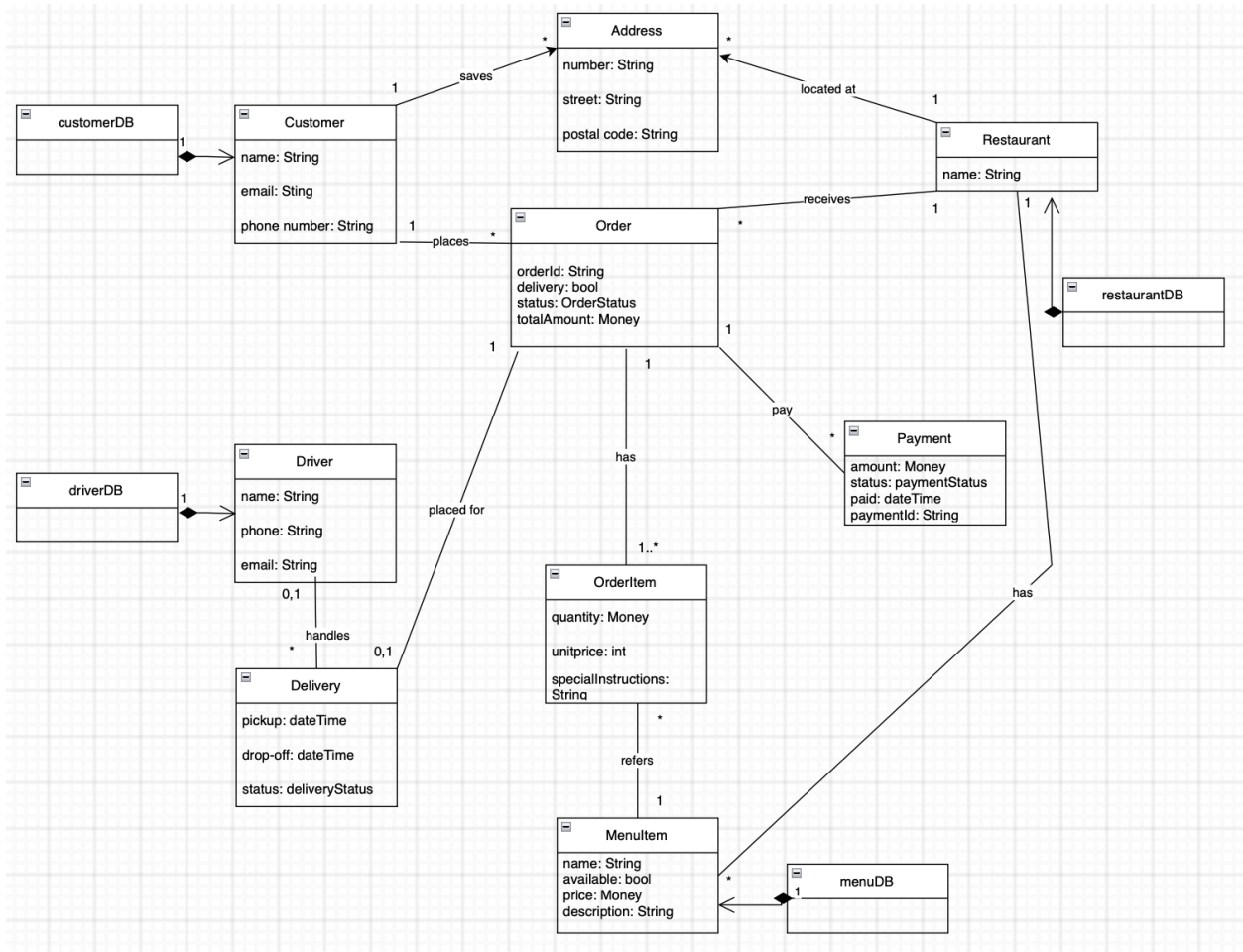
GenAI tools used

ChatGPT was used in this report to help refine the writing and explanations. It was used to review and proofread the final report, including checking grammar, formatting and clarity. It was also used for writing the references in IEEE format.

IV- Context diagram



V- Domain model



References

- [1] Django Software Foundation, “Django,” The Web Framework for Perfectionists with Deadlines. [Online]. Available: <https://www.djangoproject.com/> [Accessed: Sep. 21, 2026].
- [2] Vercel, “Next.js,” Next.js. [Online]. Available: <https://nextjs.org/>. [Accessed: Sep. 25, 2026].
- [3] Python Software Foundation, “Python.org,” Python Programming Language – Official Website. [Online]. Available: <https://www.python.org/> . [Accessed: Sep. 21, 2026].